

Generating Functions as a Counting Tool

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1 Introduction

Definition 1.1 (Ordinary Generating Function). Let $\{a_n\}_{n \geq 0}$ be a sequence of real numbers. Then the formal power series

$$A(z) = \sum_{n \geq 0} a_n z^n$$

is called the *ordinary generating function*¹ of the sequence $\{a_n\}_{n \geq 0}$.

As Herb Wilf said, a generating function of a sequence is a clothesline on which you hang all elements of the sequence. That single clothesline contains all elements of the sequence, and all information about them.

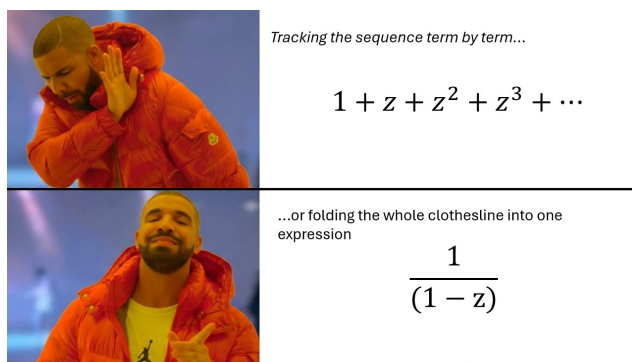


Figure 1: *Every generating function starts here.*

In this project, we will see how generating functions are used in counting problems.

¹Interestingly, ordinary generating functions are not the only kind. Another common variant which will be explored later is the *exponential generating function*, defined as $\sum_{n \geq 0} a_n \frac{z^n}{n!}$.

2 Standard Generating Function Identities

The following identities were discussed in class (Weeks 2-4). Since deriving them again is not the focus here, we collect them here for reference and use them when needed.

Sum	Closed form	Source
$\sum_{n \geq 0} z^n$	$\frac{1}{1-z}$	Geometric series
$\sum_{n \geq 0} r^n z^n$	$\frac{1}{1-rz}$	Geometric series (scaled)
$\sum_{n \geq 0} (-1)^n z^n$	$\frac{1}{1+z}$	Geometric series with $r = -1$
$\sum_{n \geq 0} (n+1)z^n$	$\frac{1}{(1-z)^2}$	Differentiate $\frac{1}{1-z}$
$\sum_{n \geq 0} n z^n$	$\frac{z}{(1-z)^2}$	Subtract $\frac{1}{1-z}$ from above
$\sum_{n \geq 0} n^2 z^n$	$\frac{z(1+z)}{(1-z)^3}$	Differentiate again
$\sum_{n \geq 0} n^3 z^n$	$\frac{z(1+4z+z^2)}{(1-z)^4}$	Differentiate again
$\sum_{n \geq 0} \binom{n+d}{d} z^n$	$\frac{1}{(1-z)^{d+1}}$	Binomial identity
$\sum_{n \geq r} \binom{n}{r} z^n$	$\frac{z^r}{(1-z)^{r+1}}$	Shifted binomial

3 Recurrence Relations and Generating Functions

To understand how generating functions help us to solve recurrence problems, let's first look at this problem:

A new coffee shop opens at SDSU and it goes viral on TikTok. Each week, the number of regular customers doubles as students tell their friends to check out the coffee shop. However, 300 customers per week stop coming because they think that baristas are mean to them. If the coffee shop opens with 500 customers in week 0, how many weekly customers will they have after n weeks?

Let a_n denote the number of weekly customers in week n . From the problem, we have $a_0 = 500$, and the recurrence relation

$$a_{n+1} = 2a_n - 300 \quad \text{for } n \geq 0.$$

Computing a few terms directly:

$$a_0 = 500, \quad a_1 = 700, \quad a_2 = 1100, \quad a_3 = 1900, \quad a_4 = 3500.$$

Although the recurrence works the problem is clear, what if we want to compute week 67? then we need to compute week 66, well for that we need to compute week 65. What we want and prefer is expression that depends only on n , so we can skip straight to answer, which is exactly what generating functions give us.

Solution

Recall that: for any constant r ,

$$\frac{1}{1-rz} = \sum_{n \geq 0} r^n z^n.$$

Let

$$A(z) = \sum_{n \geq 0} a_n z^n.$$

Multiply both sides of the recurrence by z^{n+1} and sum over all $n \geq 0$:

$$\sum_{n \geq 0} a_{n+1} z^{n+1} = \sum_{n \geq 0} 2a_n z^{n+1} - \sum_{n \geq 0} 300 z^{n+1}.$$

$$\sum_{n \geq 0} a_{n+1} z^{n+1} = a_1 z + a_2 z^2 + a_3 z^3 + \cdots = A(z) - a_0 = A(z) - 500.$$

$$\sum_{n \geq 0} 2a_n z^{n+1} = 2z \sum_{n \geq 0} a_n z^n = 2zA(z).$$

$$\sum_{n \geq 0} 300 z^{n+1} = 300z \sum_{n \geq 0} z^n = \frac{300z}{1-z}.$$

Therefore

$$A(z) - 500 = 2zA(z) - \frac{300z}{1-z}.$$

$$A(z)(1-2z) = 500 - \frac{300z}{1-z},$$

$$A(z) = \frac{500}{1-2z} - \frac{300z}{(1-z)(1-2z)}.$$

$$\frac{300z}{(1-z)(1-2z)} = \frac{C}{1-z} + \frac{D}{1-2z}.$$

$z = 1$ gives $C = -300$, and $z = \frac{1}{2}$ gives $D = 300$. So

$$A(z) = \frac{500}{1-2z} + \frac{300}{1-z} - \frac{300}{1-2z} = \frac{200}{1-2z} + \frac{300}{1-z}.$$

Expanding each term as a geometric series:

$$\frac{200}{1-2z} = \sum_{n \geq 0} 200 \cdot 2^n z^n, \quad \frac{300}{1-z} = \sum_{n \geq 0} 300 z^n.$$

Therefore

$$A(z) = \sum_{n \geq 0} (200 \cdot 2^n + 300) z^n,$$

so the closed-form formula is

$$\boxed{a_n = 200 \cdot 2^n + 300.}$$

As a sanity check, substituting into the recurrence: $2a_n - 300 = 2(200 \cdot 2^n + 300) - 300 = 200 \cdot 2^{n+1} + 300 = a_{n+1}$. ✓

Also $a_0 = 200 + 300 = 500$. ✓²

Remark 3.1 (Alternative Solution). We can get the same result from the way we did generating function recurrence in class. Rewrite the recurrence as

$$a_n = 2a_{n-1} - 300 \quad \text{for } n \geq 1,$$

separate the $n = 0$ term:

$$A(z) = a_0 + \sum_{n \geq 1} a_n z^n = 500 + \sum_{n \geq 1} a_n z^n.$$

$$A(z) = 500 + \sum_{n \geq 1} (2a_{n-1} - 300) z^n = 500 + 2 \sum_{n \geq 1} a_{n-1} z^n - 300 \sum_{n \geq 1} z^n.$$

$$\sum_{n \geq 1} a_{n-1} z^n = z \sum_{m \geq 0} a_m z^m = zA(z),$$

Also we know that

$$\sum_{n \geq 1} z^n = \frac{z}{1-z}.$$

so

$$A(z) = 500 + 2zA(z) - \frac{300z}{1-z},$$

which is the same equation we got earlier. From this point the derivation is the same and gives us $a_n = 200 \cdot 2^n + 300$. The two approaches differ is that the book multiplies the recurrence by z^{n+1} and sums over $n \geq 0$, whereas in the class we shift the index first and sum over $n \geq 1$.

²The recurrence has a fixed point at $a^* = 300$: if the shop ever has exactly 300 weekly customers, viral growth and customer loss cancel out perfectly, and the customer count stays constant forever. Above 300, growth wins and the shop goes viral. Below 300, customer loss wins: the closed-form formula becomes $a_n = (a_0 - 300) \cdot 2^n + 300$, which eventually predicts a negative customer count so sadly the shop closes.

4 Counting with Products of Generating Functions

Product rule: If $A(z) = \sum_{n \geq 0} a_n z^n$ and $B(z) = \sum_{n \geq 0} b_n z^n$, then

$$A(z) \cdot B(z) = \sum_{n \geq 0} \left(\sum_{k=0}^n a_k b_{n-k} \right) z^n.$$

We used generating functions to solve a recurrence. In this section, we will see that they handle another kind of counting problem altogether. When two generating functions $A(z) = \sum a_n z^n$ and $B(z) = \sum b_n z^n$ are multiplied, the coefficient of z^n in the product is

$$\sum_{k=0}^n a_k b_{n-k}.$$

This counts the number of ways to split an ordered list of size n into two consecutive pieces. See the following example:

You are organizing a to-do list of n tasks arranged in order of importance. You divide the list into *high-priority tasks* (the first k tasks) and *regular tasks* (the remaining $n - k$ tasks). We consider two different ways you might annotate the list.

(a) You mark **exactly 1 high-priority task as “urgent”** and **exactly 2 regular tasks as “in-progress.”** In how many ways can you annotate the list? In how many ways can you annotate the list if $n = 2$? If $n = 10$?

(b) You instead **optionally star any subset of tasks**, independently chosen in each group.³ In how many ways can you annotate the list if $n = 2$? If $n = 10$?

Solution to (a)

The first count is $\binom{k}{1}$ (choose 1 urgent task from k high-priority tasks). The second is $\binom{m}{2}$ (choose 2 in-progress tasks from m regular tasks). The corresponding generating functions are

$$A(z) = \sum_{k \geq 1} \binom{k}{1} z^k = \frac{z}{(1-z)^2}, \quad B(z) = \sum_{m \geq 2} \binom{m}{2} z^m = \frac{z^2}{(1-z)^3}.$$

The sums start at $k = 1$ and $m = 2$ because you need at least 1 task to pick 1 from it, and at least 2 tasks to pick 2.

$$F(z) = A(z) \cdot B(z) = \frac{z^3}{(1-z)^5}.$$

³For example if we have $n = 4$ and $k = 2$ then high-priority section and regular section do not constrain each other. In particular starring $\{1\}$ in the high-priority section is compatible with any starring pattern in the regular section: $\emptyset$, $\{3\}$, $\{4\}$, or $\{3, 4\}$. All $2^2 \cdot 2^2 = 16$ combinations are valid.

Using $\frac{1}{(1-z)^5} = \sum_{n \geq 0} \binom{n+4}{4} z^n$ and shifting the index by 3,

$$F(z) = \sum_{n \geq 3} \binom{n+1}{4} z^n,$$

so for $n \geq 3$:

$$f_n = \binom{n+1}{4}.$$

and for $n < 3$:

$$f_n = 0$$

For $n = 2$:

$$f_2 = 0,$$

which makes sense with only 2 tasks, the list is too short to contain both 1 high-priority and 2 regular tasks, and for $n = 10$:

$$f_{10} = \binom{11}{4} = 330.$$

Solution to (b)

Both sections of the list now have the same structure: pick any subset of the tasks. For a section of size k , there are 2^k such subsets, so both generating functions are

$$C(z) = \sum_{k \geq 0} 2^k z^k = \frac{1}{1-2z}.$$

The sum starts at $k = 0$ because the empty subset is a valid choice indeed you may choose to star no tasks at all in both sections.

$$G(z) = C(z) \cdot C(z) = \frac{1}{(1-2z)^2}.$$

Using $\frac{1}{(1-2z)^2} = \sum_{n \geq 0} (n+1) 2^n z^n$,

$$G(z) = \sum_{n \geq 0} (n+1) 2^n z^n,$$

so

$$g_n = (n+1) \cdot 2^n.$$

so for $n = 2$:

$$g_2 = 3 \cdot 4 = 12.$$

for $n = 10$:

$$g_{10} = 11 \cdot 1024 = 11264.$$

4.1 Partitions

So far the product formula has been used to split a list into two pieces and put a structure on each piece. But products of generating functions can do something much more interesting which is keeping track of how many times you use each ingredient. The classic example of this is counting partitions.

Definition 4.1 (Integer Partition). Let $a_1 \geq a_2 \geq \dots \geq a_k \geq 1$ be integers such that $a_1 + a_2 + \dots + a_k = n$. Then the sequence $(a_1, a_2, \dots, a_k)$ is called a *partition* of the integer n . The number of all partitions of n is denoted by $p(n)$. The number of partitions of n into exactly k parts is denoted by $p_k(n)$.

For example, the integer 6 has eleven partitions, shown in Figure 2 below.

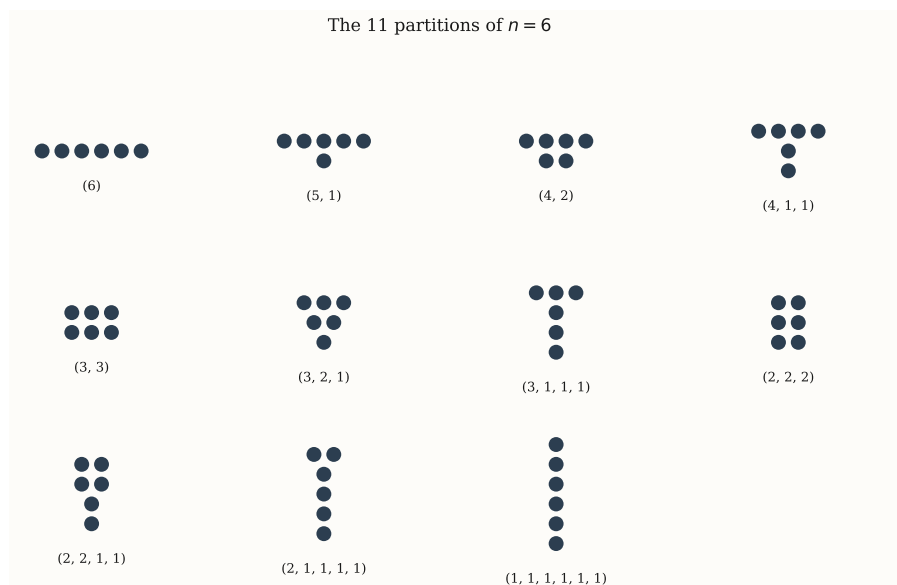


Figure 2: All eleven partitions of $n = 6$. Each row corresponds to a part a_i , with $a_1 \geq a_2 \geq \dots$.

4.2 The Generating Function for Partitions

We want a generating function $P(z) = \sum_{n \geq 0} p(n) z^n$, but we have no formula for $p(n)$ to plug in. The trick is to think about *how a partition is built*: for each part-size i , we just decide how many copies of i to use.

For each $i \geq 1$, the formal power series

$$\frac{1}{1 - z^i} = 1 + z^i + z^{2i} + z^{3i} + \dots$$

Picking the term z^{ji} means “use i exactly j times,” contributing $j \cdot i$ to the total. Multiplying across all part-sizes,

$$\prod_{i \geq 1} \frac{1}{1 - z^i} = (1 + z + z^2 + \dots)(1 + z^2 + z^4 + \dots)(1 + z^3 + z^6 + \dots) \dots,$$

each way of picking one term per factor corresponds to a partition, and the exponents add up to whatever n that partition is of. The coefficient of z^n in the product is therefore $p(n)$.

Theorem 4.2 (Generating Function for Partitions).

$$P(z) = \sum_{n \geq 0} p(n) z^n = \prod_{i \geq 1} \frac{1}{1 - z^i}.$$

We can now check Figure 2 against the theorem. Only parts of size at most 6 can appear in a partition of 6, so to extract the coefficient of z^6 from $\prod_{i \geq 1} \frac{1}{1 - z^i}$ it suffices to look at

$$(1 + z + z^2 + z^3 + z^4 + z^5 + z^6)(1 + z^2 + z^4 + z^6)(1 + z^3 + z^6)(1 + z^4)(1 + z^5)(1 + z^6).$$

⁴ Picking one term from each factor and adding the exponents gives a partition of 6:

Choice of terms	Partition
$z^6 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1$	(1, 1, 1, 1, 1, 1)
$z^4 \cdot z^2 \cdot 1 \cdot 1 \cdot 1 \cdot 1$	(2, 1, 1, 1, 1)
$z^2 \cdot z^4 \cdot 1 \cdot 1 \cdot 1 \cdot 1$	(2, 2, 1, 1)
$1 \cdot z^6 \cdot 1 \cdot 1 \cdot 1 \cdot 1$	(2, 2, 2)
$z^3 \cdot 1 \cdot z^3 \cdot 1 \cdot 1 \cdot 1$	(3, 1, 1, 1)
$z \cdot z^2 \cdot z^3 \cdot 1 \cdot 1 \cdot 1$	(3, 2, 1)
$1 \cdot 1 \cdot z^6 \cdot 1 \cdot 1 \cdot 1$	(3, 3)
$z^2 \cdot 1 \cdot 1 \cdot z^4 \cdot 1 \cdot 1$	(4, 1, 1)
$1 \cdot z^2 \cdot 1 \cdot z^4 \cdot 1 \cdot 1$	(4, 2)
$z \cdot 1 \cdot 1 \cdot 1 \cdot z^5 \cdot 1$	(5, 1)
$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot z^6$	(6)

Eleven choices, eleven partitions, $p(6) = 11$ same as the partitions shown in the figure.

4.3 Restricting What Counts as a Partition

The product $\prod_{i \geq 1} \frac{1}{1 - z^i}$ counts *all* partitions of n . But a small change to the factors counts partitions of a more restricted kind. Two restrictions are especially natural.

Distinct parts. A partition has *distinct parts* if no part is repeated, e.g. (4, 2, 1) is allowed but (3, 2, 2) is not. Let $p_d(n)$ denote the number of such partitions of n .

For each part-size i , we now have only two choices: use i exactly once, or don't use it at all. The menu shrinks accordingly:

$$1 + z^i.$$

So the generating function for distinct-part partitions is

⁴Notice that next term after z^4 would be z^8 which is already > 6 so the last three factors are just $(1 + z^4)(1 + z^5)(1 + z^6)$.

$$D(z) = \sum_{n \geq 0} p_d(n) z^n = \prod_{i \geq 1} (1 + z^i) = (1 + z)(1 + z^2)(1 + z^3)(1 + z^4)(1 + z^5) \cdots$$

Odd parts only. A partition has *odd parts* if every part is an odd integer, e.g. (5, 3, 1, 1) but not (4, 2). Let $p_{\text{odd}}(n)$ denote the number of such partitions of n .

Now we keep the original $\frac{1}{1-z^i}$ for each part-size, but we only include factors with i odd:

$$O(z) = \prod_{\substack{i \geq 1 \\ i \text{ odd}}} \frac{1}{1-z^i} = \frac{1}{1-z} \cdot \frac{1}{1-z^3} \cdot \frac{1}{1-z^5} \cdot \frac{1}{1-z^7} \cdots$$

Let's compute both for $n = 6$ by hand. The partitions of 6 into distinct parts are

$$(6), \quad (5, 1), \quad (4, 2), \quad (3, 2, 1),$$

so $p_d(6) = 4$. The partitions of 6 into odd parts are

$$(5, 1), \quad (3, 3), \quad (3, 1, 1, 1), \quad (1, 1, 1, 1, 1, 1),$$

so $p_{\text{odd}}(6) = 4$. The two counts agree. Coincidence? see the following proof:

Theorem 4.3 (Euler). *For every $n \geq 0$, $p_d(n) = p_{\text{odd}}(n)$.*

Proof. It suffices to show $D(z) = O(z)$. Using $1 + z^i = \frac{1-z^{2i}}{1-z^i}$,

$$D(z) = \prod_{i \geq 1} (1 + z^i) = \prod_{i \geq 1} \frac{1 - z^{2i}}{1 - z^i} = \frac{(1 - z^2)(1 - z^4)(1 - z^6)(1 - z^8) \cdots}{(1 - z)(1 - z^2)(1 - z^3)(1 - z^4)(1 - z^5)(1 - z^6) \cdots}.$$

Every even-indexed factor in the denominator is canceled by a numerator, leaving

$$D(z) = \frac{1}{(1 - z)(1 - z^3)(1 - z^5)(1 - z^7) \cdots} = O(z). \quad \square$$

5 Catalan Numbers

The Catalan numbers are a sequence of natural numbers that occur in various counting problems across combinatorics. The first few values are

$$1, \quad 1, \quad 2, \quad 5, \quad 14, \quad 42, \quad 132, \quad 429, \quad 1430, \quad 4862, \quad \dots$$

Named after the nineteenth-century mathematician Eugène Catalan, these numbers count over many different families of combinatorial objects. In this section we will see how they connect to generating functions.

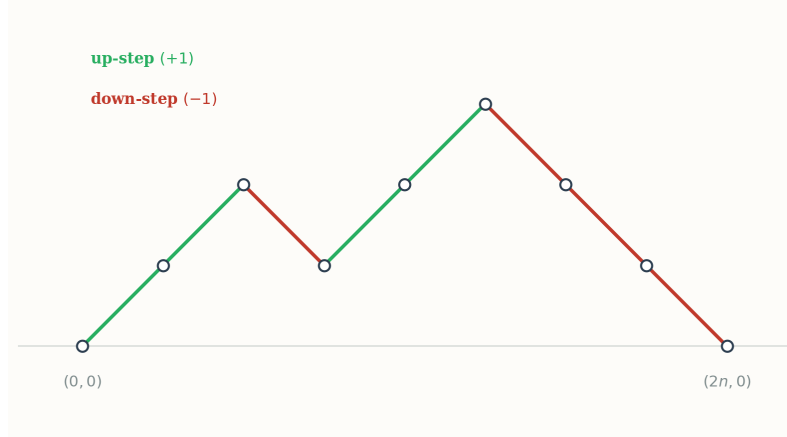


Figure 3: A Dyck path of length 8 (so $n = 4$). It uses four up-steps and four down-steps and never dips below the x -axis.

5.1 Dyck Paths

Definition 5.1 (Dyck Path). A Dyck path of length $2n$ is a lattice path from $(0, 0)$ to $(2n, 0)$ that uses only up-steps $(+1)$ and down-steps (-1) , and never goes below the x -axis.

Equivalently, a Dyck path corresponds to a sequence $b_1, b_2, \dots, b_{2n}$ of ± 1 's with n entries equal to $+1$ and n entries equal to -1 , such that every partial sum $\sum_{j=1}^k b_j$ is non-negative.

Listing all Dyck paths for small n as ± 1 sequences:

n	Dyck paths of length $2n$
1	$(+, -)$
2	$(+, +, -, -), (+, -, +, -)$
3	$(+, +, +, -, -, -), (+, +, -, +, -, -), (+, +, -, -, +, -),$ $(+, -, +, +, -, -), (+, -, +, -, +, -)$

Let c_n denote the number of Dyck paths of length $2n$, with $c_0 = 1$ for the empty path. From the table, $c_1 = 1$, $c_2 = 2$, and $c_3 = 5$ as you can see it matches the start of the Catalan sequence we saw earlier.

To set up an equation for the generating function $C(z) = \sum_{n \geq 0} c_n z^n$, we first need a recurrence for c_n . The trick is to find a way to break every Dyck path into smaller pieces in a unique way.

For any Dyck path of length $2n$ with $n \geq 1$, the path starts at height 0 and eventually ends at height 0. Look at the *first* moment after the start when the path returns to the x -axis, and call its position $2i$. (The position must be even, since the running total of ± 1 's can only equal zero after an even number of steps. In particular if we have an odd number then we cannot be at x -axis.) The value of i ranges from 1 to n , and is uniquely determined by the path.

This first return splits the path into two consecutive pieces, as shown in Figure 4.

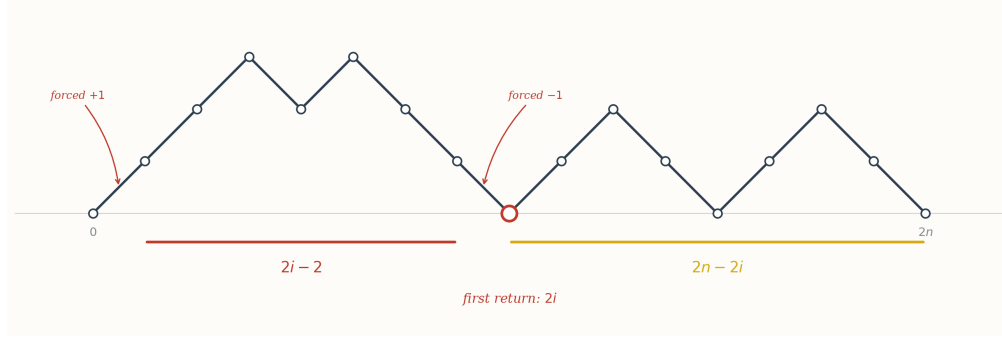


Figure 4: *Decomposition of a Dyck path at its first return to zero. The yellow piece (length $2n - 2i$) is a regular Dyck path. Notice that the first step has to be $+1$ otherwise we would go below x -axis and the step before getting to x -axis is also forced to be -1 because we can only get to x -axis from above. Stripping the forced $+1$ and -1 from the red piece yields a shorter Dyck path of length $2(i - 1) = 2i - 2$, which is our red piece*

The yellow piece. The piece from position $2i$ to $2n$ has length $2n - 2i = 2(n - i)$ and is itself a Dyck path. By definition, the number of Dyck paths of length $2(n - i)$ is c_{n-i} .

The red piece. The piece from position 0 to $2i$ has length $2i$. It starts with a forced up-step and ends with a forced down-step. After removing these two forced steps, what remains is a Dyck path of length $2(i - 1)$. Therefore the number of possible red pieces of length $2i$ is c_{i-1} . This sets up a bijection

$$\{\text{red pieces of length } 2i\} \longleftrightarrow \{\text{Dyck paths of length } 2(i - 1)\},$$

so the number of red pieces of length $2i$ is c_{i-1} using the definition.

For each fixed i , the red and yellow pieces are chosen independently, so the number of Dyck paths with first return at position $2i$ is $c_{i-1} \cdot c_{n-i}$. Summing over all possible values of i :

$$c_n = \sum_{i=1}^n c_{i-1} \cdot c_{n-i}, \quad n \geq 1.$$

i.e. at $n = 3$:

$$c_3 = c_0 c_2 + c_1 c_1 + c_2 c_0 = 1 \cdot 2 + 1 \cdot 1 + 2 \cdot 1 = 5,$$

which is same with our hand count.

The recurrence

$$c_n = \sum_{i=1}^n c_{i-1} \cdot c_{n-i}, \quad n \geq 1,$$

is exactly what the Product Rule from Section 3 was made for. we can swap the index ($j = i - 1$) to make this clearer:

$$c_n = \sum_{j=0}^{n-1} c_j \cdot c_{n-1-j}, \quad n \geq 1.$$

Now recall that

$$C(z)^2 = \sum_{n \geq 0} \left(\sum_{j=0}^n c_j c_{n-j} \right) z^n.$$

Look at the coefficient of z^{n-1} in $C(z)^2$: it is $\sum_{j=0}^{n-1} c_j c_{n-1-j}$, which is exactly the right side of our recurrence. Multiplying $C(z)^2$ by z bumps every coefficient up by one slot, so

$$z \cdot C(z)^2 = \sum_{n \geq 1} c_n z^n.$$

The right side is just $C(z)$ minus its constant term ($c_0 = 1$), so

$$\boxed{C(z) - 1 = z \cdot C(z)^2.}$$

The whole recurrence is now a single equation in $C(z)$.

Rearrange to $z \cdot C(z)^2 - C(z) + 1 = 0$, and solve the quadratic formula, and only the minus branch has constant term $c_0 = 1$ (the plus branch blows up at $z = 0$). Therefore

$$\boxed{C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}.}$$

Expanding $C(z)$ as a power series gives the coefficients c_n directly. The expansion uses a binomial theorem on $\sqrt{1 - 4z}$, a calculation that I think is too much for this project. The result is famous:

$$\boxed{c_n = \frac{1}{n+1} \binom{2n}{n}.}$$

The first few values match what we computed by hand:

$$c_0 = 1, \quad c_1 = 1, \quad c_2 = 2, \quad c_3 = 5, \quad c_4 = 14, \quad c_5 = 42, \quad \dots$$

These are the Catalan numbers we promised at the start of the section.

6 Compositions of Generating Functions

There is one more operation worth knowing: *composition*. The definition is straightforward:

Definition 6.1. Let $F(z) = \sum_{n \geq 0} f_n z^n$ be a formal power series, and let $G(z)$ be a formal power series with constant term 0. Then we define

$$F(G(z)) = \sum_{n \geq 0} f_n G(z)^n = f_0 + f_1 G(z) + f_2 G(z)^2 + f_3 G(z)^3 + \dots$$

You might wonder why definition is saying that G has constant term zero.

To see why, consider what would happen if $G(z) = 1 + z$. Then

$$G(z)^0 = 1, \quad G(z)^1 = 1 + z, \quad G(z)^2 = 1 + 2z + z^2, \quad G(z)^3 = 1 + 3z + 3z^2 + z^3, \quad \dots$$

Every single one of these has a nonzero constant term, namely 1. So the constant term of $F(G(z))$ would be

$$f_0 + f_1 + f_2 + f_3 + \cdots ,$$

an infinite sum of numbers, which formal power series simply do not allow.

If instead $G(z) = z + z^2$ (no constant term), the situation changes completely. Now

$$G(z)^0 = 1, \quad G(z)^1 = z + z^2, \quad G(z)^2 = z^2 + 2z^3 + z^4, \quad G(z)^3 = z^3 + 3z^4 + \cdots ,$$

and the lowest-degree term of $G(z)^k$ is z^k . So $G(z)^k$ contributes nothing to the coefficient of z^n whenever $k > n$, and the coefficient of z^n in $F(G(z))$ is a sum of just $n + 1$ numbers.

6.1 Counting with Composition

Composition becomes useful when it lines up with a combinatorial construction. The next two theorems give us the tools.

Theorem 6.2. *Let a_n be the number of ways to build a certain structure on an n -element set, and let us assume that $a_0 = 0$. Let h_n be the number of ways to split the set $[n]$ into an unspecified number of disjoint non-empty intervals, then build a structure of the given kind on each of these intervals. Set $h_0 = 1$. Denote $A(z) = \sum_{n \geq 0} a_n z^n$ and $H(z) = \sum_{n \geq 0} h_n z^n$. Then*

$$H(z) = \frac{1}{1 - A(z)}.$$

We do not allow empty intervals here. If we did, we could insert arbitrarily many empty intervals into any split of $[n]$, giving infinitely many ways to do it. The next theorem generalizes this by allowing a second structure on the collection of intervals itself, not just on each interval individually.

Theorem 6.3 (Compositional Formula). *Let a_n be the number of ways to build a certain structure on an n -element set, and assume $a_0 = 0$. Let b_n be the number of ways to build a second structure on an n -element set, and let $b_0 = 1$. Let g_n be the number of ways to split the set $[n]$ into an unspecified number of non-empty intervals, build a structure of the first kind on each of these intervals, and then build a structure of the second kind on the set of the intervals. Set $g_0 = 1$. Denote by $A(z)$, $B(z)$, and $G(z)$ the generating functions of the sequences $\{a_n\}$, $\{b_n\}$, and $\{g_n\}$. Then*

$$G(z) = B(A(z)).$$

The first theorem is the special case of the Compositional Formula where $B(z) = \frac{1}{1-z}$ (so $b_k = 1$ for every k , meaning there is no second structure to choose). Plugging in gives $G(z) = \frac{1}{1-A(z)}$, recovering the simpler theorem.

See the following example:

The Math 590 classroom has three whiteboard areas. The instructor, splits the class of n students into consecutive groups, then sends each group to one of the three whiteboards (multiple groups can share a whiteboard). In how many ways can this be done?

Solution. The two structures are:

- **First structure** (on each group): just the group itself. There is one way per group, so $a_k = 1$ for $k \geq 1$ and $a_0 = 0$, giving

$$A(z) = \sum_{k \geq 1} z^k = \frac{z}{1-z}.$$

- **Second structure** (on the collection of groups): assign each of the k groups to one of three whiteboards. With k groups, there are 3^k assignments, so $b_k = 3^k$ and

$$B(z) = \sum_{k \geq 0} 3^k z^k = \frac{1}{1-3z}.$$

By the Compositional Formula,

$$G(z) = B(A(z)) = \frac{1}{1-3 \cdot \frac{z}{1-z}} = \frac{1-z}{1-4z}.$$

$$G(z) = \frac{1}{1-4z} - \frac{z}{1-4z} = \sum_{n \geq 0} 4^n z^n - \sum_{n \geq 1} 4^{n-1} z^n.$$

For $n \geq 1$, the coefficient of z^n is $4^n - 4^{n-1} = 3 \cdot 4^{n-1}$.

$$\boxed{g_n = 3 \cdot 4^{n-1} \quad \text{for } n \geq 1, \quad g_0 = 1.}$$

let us do a sanity check: For $n = 2$, the class has two students $\{1, 2\}$. There are two ways to split them: as a single group $\{1, 2\}$ which can go to any of the 3 whiteboards, or as two groups $\{1\}$ and $\{2\}$ where each group independently picks 1 of 3 whiteboards (giving $3 \times 3 = 9$ choices). Total: $3 + 9 = 12$, which matches $g_2 = 3 \cdot 4 = 12$. ✓

7 Exponential Generating Functions

Remember I told you there are different generating functions and the one we saw in class isn't the only one? Well here we go. This section is going to introduce that new variant.

7.1 When OGFs Are Not the Natural Tool

Example 7.1. Let $a_0 = 1$ and $a_{n+1} = (n+1)(a_n - n + 1)$ for $n \geq 0$. Find a closed form for a_n .

Computing a few terms:

$$a_0 = 1, \quad a_1 = 2, \quad a_2 = 4, \quad a_3 = 9, \quad a_4 = 28, \quad a_5 = 125, \quad a_6 = 726, \quad \dots$$

You see how fast this sequence grows? Ordinary generating functions still exist formally, but they are not the most natural tool for this recurrence. The issue is the factor of $n + 1$ in the recurrence. If we used an ordinary generating function, this factor would lead to derivatives of the generating function.

This is exactly where exponential generating functions become useful. The $\frac{1}{n!}$ in the definition of an EGF is designed to absorb factorial-type growth, and in this example it also cancels the $(n + 1)$ factor in the recurrence. So instead of forcing OGFs to work, we use the generating function that naturally matches the structure of the recurrence.

Definition 7.2 (Exponential Generating Function). Let $\{f_n\}_{n \geq 0}$ be a sequence of real numbers. The formal power series

$$F(z) = \sum_{n \geq 0} f_n \frac{z^n}{n!}$$

is called the *exponential generating function* of $\{f_n\}_{n \geq 0}$.

The name comes from calculus. Recall the Taylor series for e^z :

$$e^z = \sum_{n \geq 0} \frac{z^n}{n!}.$$

This is exactly the EGF of the constant sequence $f_n = 1$. So the simplest EGF you can write down is the exponential function itself, hence the name. (Compare to OGFs, where the constant sequence $f_n = 1$ gives the geometric series $\frac{1}{1-z}$.)

7.2 Solving with EGFs

Remember the example where ordinary generating functions were not the most natural tool? Let's revisit it now with our new tool.

Example 7.3. Let $a_0 = 1$ and $a_{n+1} = (n + 1)(a_n - n + 1)$ for $n \geq 0$. Find a closed form for a_n .

Define the EGF

$$A(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Multiply both sides of the recurrence by $\frac{z^{n+1}}{(n+1)!}$ and sum over $n \geq 0$:

$$\sum_{n \geq 0} a_{n+1} \frac{z^{n+1}}{(n+1)!} = \sum_{n \geq 0} (n+1)(a_n - n + 1) \frac{z^{n+1}}{(n+1)!}.$$

$$\sum_{n \geq 0} a_{n+1} \frac{z^{n+1}}{(n+1)!} = \sum_{n \geq 0} (a_n - n + 1) \frac{z^{n+1}}{n!}.$$

Reindexing $m = n + 1$ on the left side: ⁵

$$\sum_{n \geq 0} a_{n+1} \frac{z^{n+1}}{(n+1)!} = \sum_{m \geq 1} a_m \frac{z^m}{m!} = A(z) - 1. \quad (\text{i})$$

$$\begin{aligned} \sum_{n \geq 0} (a_n - n + 1) \frac{z^{n+1}}{n!} &= \sum_{n \geq 0} a_n \frac{z^{n+1}}{n!} - \sum_{n \geq 0} \frac{n z^{n+1}}{n!} + \sum_{n \geq 0} \frac{z^{n+1}}{n!}. \\ \sum_{n \geq 0} a_n \frac{z^{n+1}}{n!} &= z \sum_{n \geq 0} a_n \frac{z^n}{n!} e = zA(z). \end{aligned} \quad (\text{ii})$$

Then, using $\frac{n}{n!} = \frac{1}{(n-1)!}$ and reindexing $k = n - 1$:

$$\sum_{n \geq 1} \frac{z^{n+1}}{(n-1)!} = z^2 \sum_{k \geq 0} \frac{z^k}{k!} = z^2 e^z. \quad (\text{iii})$$

$$\sum_{n \geq 0} \frac{z^{n+1}}{n!} = z \sum_{n \geq 0} \frac{z^n}{n!} = ze^z. \quad (\text{iv})$$

Substituting (i) on the left and (ii), (iii), (iv) on the right:

$$A(z) - 1 = zA(z) - z^2 e^z + ze^z.$$

Solve for $A(z)$:

$$A(z)(1 - z) = 1 - z^2 e^z + ze^z = 1 + ze^z(1 - z),$$

$$A(z) = \frac{1}{1 - z} + ze^z.$$

Now extract the coefficient of $\frac{z^n}{n!}$ from each piece.

From $\frac{1}{1-z} = \sum_{n \geq 0} z^n$: writing $z^n = n! \cdot \frac{z^n}{n!}$, the coefficient of $\frac{z^n}{n!}$ is $n!$.

From $ze^z = \sum_{m \geq 1} \frac{z^m}{(m-1)!}$: writing $\frac{1}{(m-1)!} = \frac{m}{m!}$, the coefficient of $\frac{z^m}{m!}$ is m (for $m \geq 1$, and 0 for $m = 0$).

Combining:

$$\boxed{a_n = n! + n.}$$

Checking: $a_0 = 1$, $a_1 = 2$, $a_2 = 4$, $a_3 = 9$, $a_4 = 28$, $a_5 = 125$, $a_6 = 726$ as you can see all match the recurrence. ✓

7.3 Products of Exponential Generating Functions

Just as with ordinary generating functions, the product of two exponential generating functions has a natural combinatorial meaning.

⁵notice that we have $a_0 = 1$

Lemma 7.4. Let $\{a_i\}$ and $\{b_k\}$ be two sequences, and let

$$A(z) = \sum_{i \geq 0} a_i \frac{z^i}{i!}, \quad B(z) = \sum_{k \geq 0} b_k \frac{z^k}{k!}$$

be their exponential generating functions. Define $c_n = \sum_{i=0}^n \binom{n}{i} a_i b_{n-i}$, and let $C(z)$ be the exponential generating function of $\{c_n\}$. Then

$$A(z) \cdot B(z) = C(z).$$

In other words, the coefficient of $\frac{z^n}{n!}$ in $A(z)B(z)$ is $c_n = \sum_{i=0}^n \binom{n}{i} a_i b_{n-i}$.

In case you are wondering where the binomial coefficient comes from, look at a general term in the product $A(z) \cdot B(z)$:

$$a_i \frac{z^i}{i!} \cdot b_j \frac{z^j}{j!} = a_i b_j \cdot \frac{z^{i+j}}{i! j!} = a_i b_j \cdot \frac{(i+j)!}{i! j!} \cdot \frac{z^{i+j}}{(i+j)!} = a_i b_j \binom{i+j}{i} \cdot \frac{z^{i+j}}{(i+j)!}.$$

This contributes to $\frac{z^n}{n!}$ exactly when $i+j = n$, which gives the lemma's formula for c_n .

Observation. Multiplying $\frac{1}{1-z}$ by itself gives $\frac{1}{(1-z)^2} = \sum (n+1)z^n$, which counts the $n+1$ ways to cut a sequence in two. But multiplying e^z by itself gives $e^{2z} = \sum 2^n \frac{z^n}{n!}$, which counts the 2^n ways to send each of n labels independently to one of two groups.⁶

Example: Math 590 Whiteboard Practice with Markers

Recall the whiteboard example from Section 6: a list of n students gets split into consecutive groups, and each group is sent to one of three whiteboards. There the students were positions in a fixed roster, and the OGF Product Rule handled the consecutive-cut split.

Now consider a closely related problem where the students are individuals rather than positions.

The Math 590 class has n students. Instructor splits the class into two possibly empty labeled groups: a “working group” and an “observing group.” Each group lines up at its own whiteboard. Then each member of the working group picks one of three marker colors (red, blue, or green) to write with; the observing group just watches. In how many ways can this be done?

Solution. The two structures are:

- **Working group** (size k): the students form a line ($k!$ orderings) and each picks a marker color (3^k choices). So $a_k = k! \cdot 3^k$, giving

$$A(z) = \sum_{k \geq 0} k! \cdot 3^k \cdot \frac{z^k}{k!} = \sum_{k \geq 0} 3^k z^k = \frac{1}{1-3z}.$$

⁶For example, with $n = 3$ labels $\{1, 2, 3\}$: the OGF count gives 4 (cut before 1, after 1, after 2, or after 3), while the EGF count gives $2^3 = 8$ (each label independently goes left or right, including splits like $\{1, 3\}$ vs $\{2\}$ that the OGF version doesn't allow).

- **Observing group** (size m): the students form a line, no other choice. So $b_m = m!$, giving

$$B(z) = \sum_{m \geq 0} m! \cdot \frac{z^m}{m!} = \sum_{m \geq 0} z^m = \frac{1}{1-z}.$$

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By the EGF Product Rule:

$$C(z) = A(z) \cdot B(z) = \frac{1}{(1-3z)(1-z)}.$$

$$\frac{1}{(1-3z)(1-z)} = \frac{A}{1-3z} + \frac{B}{1-z}.$$

Solving partial fractions gives us $A = 3/2$ and $B = -1/2$:

$$C(z) = \frac{3/2}{1-3z} - \frac{1/2}{1-z} = \sum_{n \geq 0} \frac{3^{n+1} - 1}{2} z^n.$$

To convert this OGF-style coefficient to an EGF coefficient, multiply by $n!$:

$$c_n = n! \cdot \frac{3^{n+1} - 1}{2}.$$

Doing a sanity check: For $n = 1$, the formula gives $c_1 = 1 \cdot \frac{9-1}{2} = 4$. Counting by hand: the single student either goes to the working group (with one of 3 marker colors) or the observing group, totaling $3 + 1 = 4$ ways. ✓

7.4 The Exponential Formula

The composition story for OGFs had a clean form: splitting $[n]$ into consecutive intervals and structuring each gave $H(z) = \frac{1}{1-A(z)}$. For EGFs, the split is into unordered subsets instead, and the formula uses the exponential function in place of the geometric series.

Theorem 7.5 (The Exponential Formula). *Let a_n be the number of ways to build a structure on an n -element set, with $a_0 = 0$. Let h_n be the number of ways to partition $[n]$ into an unspecified number of non-empty unordered subsets and build a structure of the given kind on each subset. Set $h_0 = 1$. If $A(z)$ and $H(z)$ are the EGFs of $\{a_n\}$ and $\{h_n\}$, then*

$$H(z) = e^{A(z)}.$$

By the EGF Product Rule, the EGF for “partition $[n]$ into k ordered non-empty subsets and structure each” is $A(z)^k$. But we want *unordered* subsets, and each unordered partition into k blocks corresponds to $k!$ ordered versions (one for each

⁷Notice how cleanly the factorials cancel.

way to order the blocks). Dividing by $k!$ corrects the over-count, so $\frac{A(z)^k}{k!}$ is the EGF for “partition into exactly k unordered blocks, structure each.” Summing over all k :

$$H(z) = \sum_{k \geq 0} \frac{A(z)^k}{k!} = e^{A(z)},$$

since the right side is the Taylor series for the exponential.

The two “many pieces” theorems sit side by side:

	OGF	EGF
Splitting rule	Consecutive intervals	Unordered subsets
Formula	$H(z) = \frac{1}{1 - A(z)}$	$H(z) = e^{A(z)}$

Before the example, we need a small identity. Integrating the geometric series term-by-term:

$$\int \frac{1}{1-z} dz = \int \sum_{n \geq 0} z^n dz = \sum_{n \geq 0} \frac{z^{n+1}}{n+1} = \sum_{k \geq 1} \frac{z^k}{k}.$$

The same integral evaluates to $\ln\left(\frac{1}{1-z}\right)$ in calculus, so

$$\sum_{k \geq 1} \frac{z^k}{k} = \ln\left(\frac{1}{1-z}\right).$$

Now let’s work on this example:

In how many ways can n people be arranged into groups, where each group sits at its own circular table?

Solution. The structure on each group is “seat k people at a circular table.” If they sat in a line, there would be $k!$ orderings, but rotating a circular seating gives the same arrangement, and there are k rotations. So $a_k = (k-1)!$ for $k \geq 1$, with $a_0 = 0$.

The EGF is

$$A(z) = \sum_{k \geq 1} (k-1)! \cdot \frac{z^k}{k!} = \sum_{k \geq 1} \frac{z^k}{k} = \ln\left(\frac{1}{1-z}\right).$$

By the Exponential Formula,

$$H(z) = e^{A(z)} = e^{\ln \frac{1}{1-z}} = \frac{1}{1-z}.$$

The exponential and logarithm cancel cleanly. Extracting the coefficient of $\frac{z^n}{n!}$:

$$\frac{1}{1-z} = \sum_{n \geq 0} z^n = \sum_{n \geq 0} n! \cdot \frac{z^n}{n!},$$

so $h_n = n!$.

7.5 The EGF Compositional Formula

Just as the OGF compositional formula generalized, the Exponential Formula has its own generalization.

Theorem 7.6 (Compositional Formula, Exponential Version). *Let a_n be the number of ways to build a certain structure on an n -element set, with $a_0 = 0$. Let b_n be the number of ways to build a second structure on an n -element set, with $b_0 = 1$. Let g_n be the number of ways to partition $[n]$ into an unspecified number of non-empty subsets, build a structure of the first kind on each subset, and then build a structure of the second kind on the set of subsets. Denote by $A(z)$, $B(z)$, and $G(z)$ the EGFs of $\{a_n\}$, $\{b_n\}$, and $\{g_n\}$. Then*

$$G(z) = B(A(z)).$$

The idea is the same as the Exponential Formula, with the second structure layered on top. For a fixed number of subsets k , the EGF for “partition into k unordered subsets, structure each subset individually, and structure the collection of subsets as a whole” is $b_k \cdot \frac{A(z)^k}{k!}$. Summing over all k :

$$G(z) = \sum_{k \geq 0} b_k \frac{A(z)^k}{k!} = B(A(z)).$$

8

All theorems we have seen fit cleanly into one table:

Split type	No second structure	Second structure
Consecutive intervals (OGF)	$\frac{1}{1 - A(z)}$	$B(A(z))$
Unordered subsets (EGF)	$e^{A(z)}$	$B(A(z))$

The whiteboard problem that we keep getting back to used OGFs because the list was in a fixed order. If instead the students are individuals being freely sorted, and each group forms a line at its whiteboard, the same setup becomes EGF question.

The Math 590 class has n students. Instructor splits the class into non-empty unordered groups, each group forms a line at its whiteboard, and each group is assigned to one of three whiteboards. In how many ways can this be done?

Solution.

First structure (on each group of k students): form a line, $k!$ ways. So $a_k = k!$ for $k \geq 1$, and $a_0 = 0$. Since we are using an exponential generating function,

$$A(z) = \sum_{k \geq 1} k! \cdot \frac{z^k}{k!} = \sum_{k \geq 1} z^k = \frac{z}{1 - z}.$$

⁸Note that for $k = 0$ we get 1.

Second structure (on the collection of k groups): each group picks one of three whiteboards, so there are 3^k choices. Thus $b_k = 3^k$. Since this is also an exponential generating function, we have

$$B(z) = \sum_{k \geq 0} 3^k \frac{z^k}{k!} = e^{3z}.$$

By the EGF Compositional Formula,

$$G(z) = B(A(z)) = e^{3A(z)} = e^{\frac{3z}{1-z}}.$$

Therefore,

$$g_n = n! [z^n] e^{\frac{3z}{1-z}}$$

⁹ for $n \geq 0$.

As a sanity check, let $n = 2$. If both students are in one group, then the group can be lined up in $2!$ ways and assigned to one of three whiteboards, giving

$$2! \cdot 3 = 6$$

choices. If the students are split into two singleton groups, then each group chooses one of three whiteboards, giving

$$3^2 = 9$$

choices. So the total is

$$6 + 9 = 15.$$

Now let's verify this matches the generating function. We extract $[z^2] e^{\frac{3z}{1-z}}$ directly. Expanding the exponential,

$$e^{\frac{3z}{1-z}} = \sum_{k \geq 0} \frac{1}{k!} \left(\frac{3z}{1-z} \right)^k.$$

Only the terms with $k = 0, 1, 2$ can contribute to $[z^2]$, since $\left(\frac{3z}{1-z} \right)^k$ has lowest-degree term z^k , so any $k \geq 3$ starts at z^3 or higher.

- $k = 0$: the term is 1, contributing 0 to $[z^2]$.
- $k = 1$: $\frac{3z}{1-z} = 3z + 3z^2 + 3z^3 + \dots$, contributing 3 to $[z^2]$.
- $k = 2$: $\frac{1}{2} \left(\frac{3z}{1-z} \right)^2 = \frac{9z^2}{2(1-z)^2} = \frac{9}{2}(z^2 + 2z^3 + \dots)$, contributing $\frac{9}{2}$ to $[z^2]$.

Therefore

$$[z^2] e^{\frac{3z}{1-z}} = 3 + \frac{9}{2} = \frac{15}{2},$$

and

$$g_2 = 2! \cdot [z^2] e^{\frac{3z}{1-z}} = 2! \cdot \frac{15}{2} = 15. \quad \checkmark$$

The two methods agree.

⁹Unlike the earlier examples, this is not a fully simplified closed form in particular it is a coefficient-extraction formula. It is quite out of scope to derive it.

8 Conclusion

We saw how generating functions, the same tool that was used to prove Ehrhart's theorem in our discrete geometry class, are also used in combinatorics to count.

References

- [1] Miklós Bóna, *A Walk Through Combinatorics: An Introduction to Enumeration and Graph Theory*, 3rd edition, World Scientific, 2011.